2022/23 Extended Essay in Theatre

Title: The Role of Lighting Within Theatre

Research Question: How does Lighting within the National Theatre's Frankenstein affect an Audience's perception of the play?

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Introduction

"Even with a great script and exceptional actors, poor stage lighting could end up ruining everything, going as far as to change the entire feel of a production" (DIGITTO MEDIA, 2019). It is because of this that I see the importance of lighting choices in theatre, perhaps on the same level as the script or actors on stage as lighting equally affects the way its spectator perceives the overall production. This has led me to question what lighting choices are considered when designing lighting for a play, and how exactly does this affect the audience's perception of one. In Stanley McCandless A Method of Lighting the Stage, McCandless separates the way lighting is used to illuminate the stage into intensity, colour, distribution, and control. However, as McCandless states "*Control of the light illuminating the acting area, as for all other lighting, involves the control of intensity, colour, distribution, and changes.*" (McCandless, 1947, p.60), I believe that the factor of Control will naturally have to be analysed in combination with other factors, and instead wish to place more emphasis on the type of lighting instrument used. Lighting Instruments have a role in determining what type of effect can be created in the first place, as each instrument is unique in the ways it illuminates the stage and the way they are deployed making them an integral factor in examining and analysing lighting choices. Taking all the above into account, this essay aims to examine how lighting choices based on Colour, Distribution and Intensity, as well as the type of lighting instruments used may play a role in enhancing a scene towards an audience.

However as important as defining how lighting choices are in affecting the spectators view of the play is how an audience perceives a play itself. According to Dr. Eric W Trummbull, theatre may appeal to an audience through Sensory Stimulation through

light and sound, the story and character in a play, the artistic value of a play, as well as the “intellectual value” or the theme of the play (Trumbull, 2007). Through these definitions is it possible to analyse how the lighting choices above affect said methods the audience perceives a play, and I believe the best way to demonstrate this view would be to use examples from a play where lighting choices excel in this regard, leading me to the National Theatre’s Frankenstein as the subject of my analysis. I have chosen this play, not only due to the suggestion where lighting has affected a spectators experience greatly given the praise lighting has in reviews, but also due to the fact I can be ensured that lighting is used to its fullest potential through the play having received multiple awards in lighting design. While this essay will not be a critique of Frankenstein’s lighting, it will use scenes where lighting influences an audience’s perception in order to answer **How does Lighting within the National Theatre’s Frankenstein affect an Audience’s perception of the play?**

1.Type of Lighting Instruments

There are many different types of instruments for stage lighting, each lantern usually being unique in the way they produce an effect or illuminate the stage (Illuminated Integration, 2021). While instruments in stage lighting are most referred to as the different lanterns used to produce light onto a stage, in reality these are only primary lighting instruments, with McCandless stating that secondary lighting instruments do not even need to take up the form of lanterns themselves. "*Scenery, costumes, and even makeup can be called secondary lighting instruments due to light not being visible until it strikes some surface so that it can be reflected to the eye*." (Mccandless, p.15) Both Primary and Secondary lighting instruments compliment each other, thus achieving certain effects that are only possible through both instruments working to enhance not just the target of illumination, but the features of the specific instruments themselves. Using the right lighting instrument choices, it is possible to immerse the audience into the play by bringing out certain themes and motifs of the play.

Frankenstein, in its use of lanterns prioritised tungsten and filament lanterns over LED due to their nature of being able to achieve the "warm" temperatures and be able to go up to high intensities. As stated by Bruno Poet the lighting designer, tungsten was used "*both for its warmth, and because it better suggested the industrial age, the time period of the show.*" (Halliday, 2011) suggesting the role said lanterns had on core themes of the play such as that of the industrial revolution. These features were able to affect how the audience perceived the play, with an example being seen in the use of filament light bulbs hanging above the ceiling of the stage. Said light bulbs were hung close enough to the audience and were

individually controlled such that the features of the light bulbs could be enhanced to its fullest potential through not only stimulating the senses of the audience, but also by delivering core themes of Frankenstein. This could be seen most notably in the first scene of the play, where the bulbs were able to be set to a high level of intensity such that not only was the audience blinded by the bulbs, but also were able to feel the heat coming off the light bulbs (figure 1) (Wiegand, 2020).



figure 1. Scene 1 of National Theatre Live at Home's Frankenstein (with Benedict Cumberbatch as the creature). Pictures taken from 04:28 and 08:09 respectively.

In addition to shocking their senses, the light bulbs were controlled to create a pattern reminiscent of a wave of light, being fed into a set piece representing a "womb" where the creature would be birthed from. Playmakers hoped that through this display were the audience able to relate to the creature on an emotional level by

immersing them in the foreignness of electricity, and thus grasp themes of galvanism of Frankenstein's story (etnow, 2011). The same bulbs are used differently in Scene 6, where instead of being set to a low instead of high intensity, and controlled to flicker individually thus creating the illusion of stars (figure 2).



figure 2. Scene 6 of National Theatre Live at Home's Frankenstein (with Benedict Cumberbatch as the creature). Picture taken from 16:40

The effect allows the audience to share the wonder of discovering nature with the character on stage, while at the same time reinforcing his humanity by suggesting to the audience that the creature could have these feelings and emotions. The use of the bulbs in the scenes highlights how each unique primary lighting instrument can not only create their own individual effect within theatre, but also demonstrate how the unique effects created by these lanterns cause the audience to further immerse themselves into the characterisation of those of stage. They stimulate the senses of the audience through sight and touch which may further allow them an understanding of characters emotions and core themes of the play.

The use of Secondary lighting instruments to enhance lanterns can also be seen within Frankenstein. The play utilises many traditional lanterns such as Fresnels, and

to highlight the features of these lanterns, parts of the set are used to compliment them. This can be seen In Scenes 6-8 where they are used to create the effect of sunrise and sunlight (figure 3,4).



figure 3. Scene 6 of National Theatre Live at Home's Frankenstein (with Benedict Cumberbatch as the creature). Picture taken from 17:30



figure 4. Scene 7 of National Theatre Live at Home's Frankenstein (with Benedict Cumberbatch as the creature). Picture taken from 19:30

Both effects are created using McCandless's method of imitating sunlight, by directing the rays of lights representing the sun and directing them onto the acting area as well as a piece of scenery on stage (McCandless, p112). Fresnels give off a

more “natural” beam of light due to them using a filament based lamp, being able to create beams of light that are relatively soft and diffused (Salzberg & Kupferman, 2009-2018), and by reflecting said diffused beam off a reflective surface on the cyclorama is the audience able to view this effect in the form of sunrise. The reflectable cloth acts as a secondary lighting instrument to compliment the beams of the Fresnels, reflecting off beams that are just soft enough to illuminate the actor and keep the stage dim. At the same time, the piece of reflective cloth is illuminated and highlighted for the audience to see, forming the shape of the sun at dawn. Following the sunrise, Fresnels placed on top of the stage are shined downwards towards the stage at a higher intensity with the fixtures of the instruments positioned such that the diffused beams would form a pattern made to look like rays of the sun coming down. To further emphasise this, the scenery of green grass is illuminated easily allowing the audience to link the scene to its corresponding themes of the discovery and beauty of nature. The entire sequence is made to appear magical to the audience as to simulate what goes on in the creature's mind, with choices regarding the instruments and their position painting a cartoonish picture, allowing the audience to see his childlike character through how he is able to perceive nature as a sort of magical discovery. From this, it could be determined that the importance of secondary lighting instruments complimenting lanterns are not only to create the emotional effect of a play in its audience, but also deliver core themes within the play as seen by how the audience is able to humanise the creature by perceiving its ability to discover and learn. The effectiveness of Frankenstein lighting choices in this scene can be seen through its reviews, with the New York times describing the sequence as a “gorgeous vignette” when the lighting is coupled with sound, going on to say how they are “vital in making the play’s illusions soar” demonstrating how

lighting instruments are critical in how the audience may perceive a play (Brantley, 2011).

2. Colour of Lanterns

The Colour of Lanterns are also another factor integral towards the audience's perception of a play due to how colour has a psychological effect on humans. They are able to influence our overall mood and emotions, furthermore prompting an emotional response to specific colours; some common examples being the colour red for anger, or black/dim light for power and hostility (Illuminated Integration, 2022). McCandleless portrays colours in lanterns as mostly a supplement towards secondary lighting instruments such as makeup and costumes, stating how they may enhance or define certain character traits by illuminating said secondary lighting instruments in a certain colour scheme (McCandless, p.50). These colour schemes may vary between shows or even scenes, with some believing it more appropriate to use a monochromatic colour scheme to emphasise a single colour, while some others may even combine colours in order to create certain effects or make more complex colour schemes that allow an audience to make links towards certain moods trying to be brought out by the scene. (Illuminated Integration, 2022) However, it is not only because of core themes or the illumination of the stage do designers place great emphasis on colour. It is absolutely necessary to completely control the amount and type of colour being shone onto the stage to make the play a pleasant viewing experience towards the audience due to the inappropriate use of colour leading to the overstimulation of the senses resulting in negative effects such as colour fatigue. (Illuminated Integration, 2022)

An example of a monochromatic colour scheme used to enhance an audience's viewing experience of Frankenstein can be seen in Scene 24 (figure 5), where the colour scheme is used not only to support secondary lighting instruments on stage,

but also to act as motifs in the scene in order to make clear certain themes towards the audience.



figure 5. Scene 24 of National Theatre Live at Home's Frankenstein (with Benedict Cumberbatch as the creature). Picture taken from 56:26

On the surface level, an examination of the lighting scheme suggests that lights are used to create the setting of the scene of a snowy mountain, using naturally cold white lights to illuminate the entire stage area in a colour wash while also emphasising the smoke from haze machines to simulate a snowy fog surrounding the stage. However, the monochromatic colour scheme of white allows for the actors' facial expressions and costumes to be highlighted clearly due to the nature of its colour, allowing the audience to link with the conflict on stage emotionally, as such a colour scheme would allow actors to make full use of the stage and also perform to their fullest extent as the lighting not only does not restrict the performance, but also supplements it by providing a higher degree of visibility. In addition to visibility, the colour scheme may also supplement the themes of the scene by acting as a motif. The scene holds themes of divinity, solitude, as well as hatred brought out by the dialogue between the two showing great awe towards each other. From it holding

connotations of heaven, spirituality, and faith (Bourn, 2010) The colour white therefore becomes a motif of the divine in the scene, allowing the audience to feel the inner emotions of the characters as if they were meeting the divine. Alternatively, the colour white may also be used to represent the character traits of solitude and hatred towards each other as seen by how white is repetitively used in other scenes where these two themes are being brought out such as in Scene 5 when the creature wanders the streets of Ingolstadt and beaten (figure 6), or in the final scene of the play where blue light is used in combination with white to represent the loneliness of victor and the creature (figure 7).



figure 6. Scene 5 of National Theatre Live at Home's Frankenstein (with Benedict Cumberbatch as the creature). Picture taken from 16:03



figure 7. Scene 30 of National Theatre Live at Home's Frankenstein (with Benedict Cumberbatch as the creature). Picture taken from 1:50:28

Audience Reviews of Frankenstein capture how these lighting choice were able to successfully deliver the emotional and intellectual themes of the play, noting how they felt a sense of "both characters despising yet admiring each other" thanks to "Bruno Poet's gothic lighting" (Clarerendon, 2020).

As stated by Lee Watson in the Educational Theatre Journal, "saturated colours when used and mixed in combination with other coloured lights may be used together with each other to create an overall colour that is richer than the use of a single colour." (Watson, 1958, p.257) Apart from lighting schemes of a monochromatic nature, Frankenstein is also able to expertly demonstrate how colours are mixed to complement each other, either to soften them such that they seem natural towards an audience and prevent their eyes from getting overstimulated, or to enhance the colour scheme to make the effect created by the lighting richer. The preset and beginning scene of Frankenstein are made to look as if it were taking place inside a "womb" (Bloomsbury Media, p.6), thus allowing the audience to humanise the creature from the start as they view his birth (figure 8).

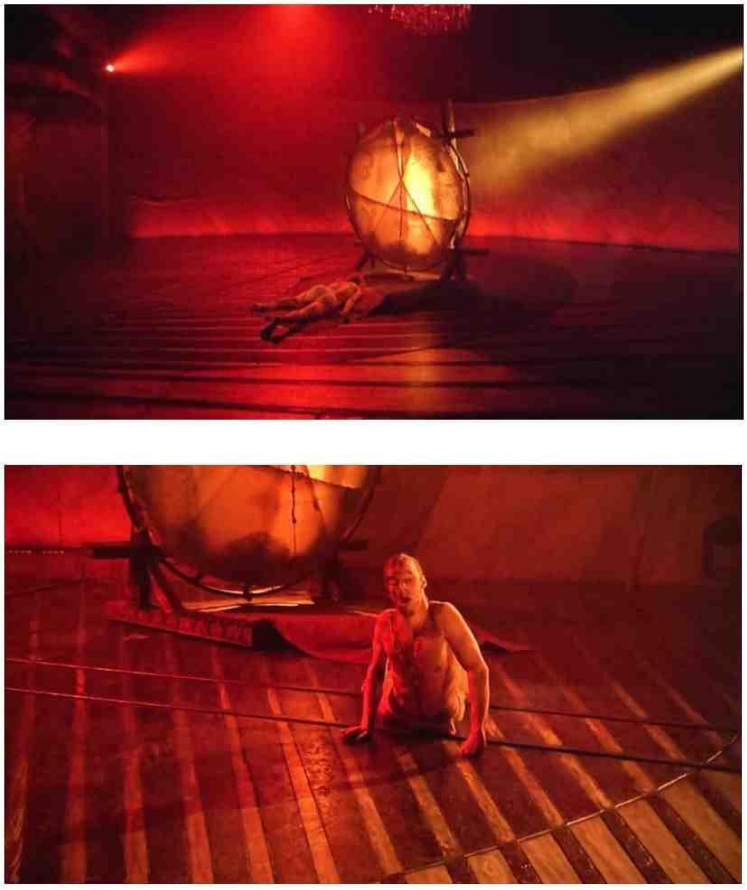


figure 8. Scene 1 of National Theatre Live at Home's Frankenstein (with Benedict Cumberbatch as the creature). Pictures taken from 05:29 and 07:07 respectively

The colour red is obviously used in the scene in what looks as if it were a colour wash, but as red is a more saturated colour, it would not only blend too much with the actors costume or makeup using red, but also make the play look artificial towards the audience if used alone at a higher quantity, being unlikely to evoke the emotions or themes playmakers were attempting to deliver. Therefore, lanterns containing red were intentionally set at a lower intensity, while other lanterns containing yellow and orange were added to illuminate the actor portraying the creature in order to separate the backdrop of red from the acting area. Even though yellow and orange are relatively eye irritable colours (Morton), red in this scene

mixes with it to prevent it from making the audience feel unpleasant, with these two colours even working to supplement other elements of theatre on stage. Through their addition, the patterns on the creature's costume are better perceived by the audience due to the colours yellow and orange being able to highlight the features of the costume or makeup by contrasting the red while at the same time not removing the reddish tone that makes the scene look as if it were inside a womb due to them being adjacent colours to red. The yellow and orange combining at the same time mimics light coming from outside the skin, further allowing for the audience to view it as one and thus allowing playmakers to deliver a core message. The importance of colour mixing is highlighted within this scene in Frankenstein, demonstrating how mixing adjacent colours as well as using colour mixing schemes in lighting allows an audience to perceive a play better.

3. Distribution and Intensity of Lights

The Distribution and intensity of lights refer to how light is allocated onto the stage in terms of the position a beam of light is as well as how bright it should be. Stage lighting is not often created with just a simple colour wash setting and designers usually create their light plans in order to achieve “selective visibility”, in which light leads an audience’s focus to a particular part of the stage (Salzberg & Kupferman, 2009-2018). By focusing the audience’s view on certain elements, light is able to draw focus onto specific characters or set pieces thus at the same time thus playing a supporting role for other theatrical elements to stimulate the audience’s senses. While this may simply sound like distribution and intensity’s role is limited towards illuminating the stage, it is incorrect to believe it plays a role only in this regard. As stated by McCandless, “*The distribution of light over the acting area should give what might be called dramatic visibility. Some areas are more important than others and sometimes the best visibility from a dramatic point of view is that which does not allow too clear a delineation of form.*” (McCandless, p.52) Just like the type of instruments and the colour of lanterns, Distribution and Intensity of lights can be altered such as to build the atmosphere of the scene or convey emotions of characters. McCandless suggests that instead of simply just focusing on illuminating the actor and set, it is important to distribute or set lighting in a way which allows for the audience to understand what is happening from a dramatic perspective with it being alright to sacrifice some degree of visibility in order to portray the core of the scene to the audience.

It can be seen how Frankenstein balances selective and dramatic visibility in its lighting choices. As stated in Section 2 of this essay, Scene 24 uses a

monochromatic colour scheme of white in order to bring out its themes towards the audience (figure 5). Equally as important however is the way in which light is distributed, with designers intentionally focusing a line of lights onto the smoke from haze machines thus effectively cutting the acting area of the stage by half, while at the same time still preserving the atmosphere of the setting by directing low intensity white lights towards the backdrop, utilising the transparency of the smoke to make it seem as if the audience were on top of a boundless mountain. In order to ensure that the audience's attention is still kept on the actors, other lights at a high intensity are aimed towards the front half of the stage, diverting audience attention towards the current acting area of the scene furthermore allowing a greater visibility of the actors faces, drawing the audience into the emotional conflict the two characters are having in the scene and thus appreciate the actors characterisation. The scene demonstrates the use of distribution and intensity to balance out illumination and dramatic visibility towards an audience, showing how lighting choices may play a role in storytelling in addition to just allowing the audience to perceive the play.

Frankenstein also holds examples in which the illumination of a scene is sacrificed rather than balanced in order to increase the dramatic visibility of a play. This is most evidently seen in how the distribution and the intensity of lights is used to intentionally discomfort the audience at times allowing the audience to relate with the creature on an emotional level in terms of the foreignness of the industrial revolution. In Scene 4 of the play, high intensity lights are placed on top of a steam locomotive and shined directly at the audience in combination with the use of pyrotechnics (figure 9) (Bloomsbury Media, p.10).

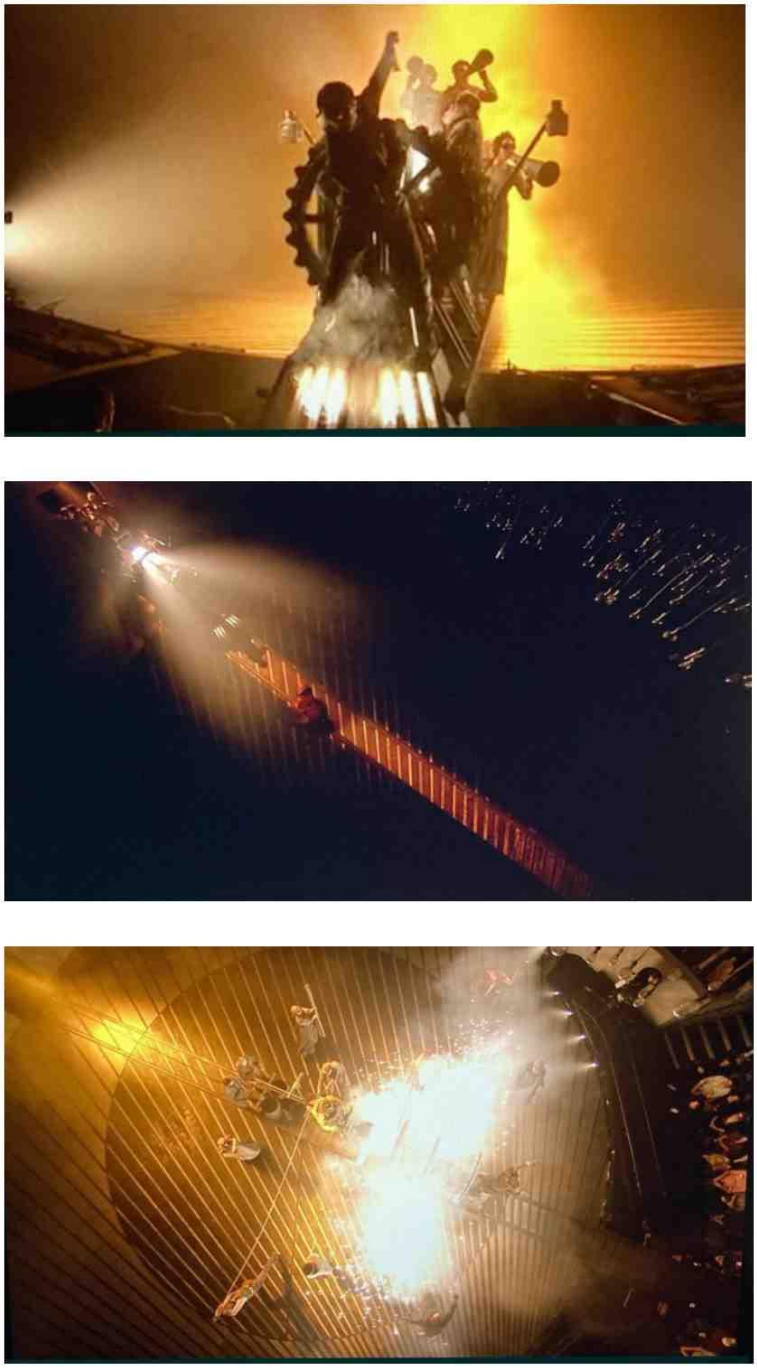


figure 9. Scene 4 of National Theatre Live at Home's Frankenstein (with Benedict Cumberbatch as the creature). Pictures taken from 13:16, 13:28 and 14:02 respectively

While said effect could cause minor discomfort through overstimulating the audience's sight, this effect allows them to relate to the creature's feelings towards said foreign elements by experiencing the overstimulation of senses that the creature

felt at the time. While it could be argued that such an effect could work negatively towards the audience by causing fatigue in the eyes from overstimulation, lighting designers ensured to control the amount of discomfort the audience received by placing the lantern behind parts of a locomotive vehicle, with such control of lighting allowing Frankenstein to be successful in delivering their themes, as seen by reviews stating how the sequence has added an extra sense of drama to the scene (theatretechgeek, 2020). Similarly, light bulbs were intentionally placed at a low level close to the audience (Wiegand, 2020) and used in the opening scene of the play in order to overstimulate the audience's senses through producing heat that the audience felt and blinding them when set at a high intensity level (figure 1). While these effects were achieved due to the nature of the light bulbs themselves, the effect was emphasised by the setting on their placement and intensity making them an integral part in achieving the effect in the first place. Like the example in Scene 4, reviews indicate that although the audience felt slight discomfort in these moments of the show were they able to grasp messages from the play on the industrial age and electricity (etnow, 2011), Frankenstein thus demonstrating how lighting choices prioritising a plays message and dramatic features would be able to allow the playgoer a better experience and understanding of the show.

Conclusion

Even with a great script and exceptional actors, poor stage lighting could end up ruining everything, going as far as to change the entire feel of a production (DIGITTO MEDIA, 2019). This is a sentiment that I personally strongly agree with, and believe that every decision made on illuminating a stage should be taken into account such that not only lighting elements, but other theatrical elements be brought out to their fullest potential. While the three main factors I believe contributing towards an audience's perception– Types of instruments, colour, as well as distribution and intensity are examined separately throughout this essay, it is clear how each factor works to compliment each other, especially when analysed against an example of how they are used. While this essay cannot determine which factors in lighting are more or less important in affecting audience perception, I believe that this can be a question for future discussion, in which these three points are examined to determine which is prioritised more in lighting design. I have already begun this exploration by taking on light design work in Hong Kong Sweet and Sour Productions Christmas Carol School performances, in which I would have a chance to use the theories and practices I have learnt from this essay, and personally experience how a lighting designer utilises them in a practical environment.

There are a number of factors that make this research paper limited. The first would be how I, the writer of this paper, have not studied theatre in any capacity before with my knowledge of lighting or even theatre in general being limited to my own practical knowledge working in a variety of backstage roles. While I believe that said practical knowledge is able to counteract my lack of knowledge, there is no denying that I may have overlooked obviously relevant theories or aspects of theatre that I failed to

realise. Regardless of this, I do believe that I have to the best of my ability attempted to research and acquire knowledge on the field of lighting such that this research paper may be valuable towards its views on the effects of lighting on an audience. Another limitation of the paper would be how its scope is limited to research on a single play and is heavily reliant on the theories and methods of Lighting design from Stanley McCandless. While it is a fact that this research by analysing a single play in depth has allowed me to better understand factors influencing an audience's perception of theatre and that McCandless is renowned in the world of stage lighting, there may be other plays or other modern theorists that take a different approach into examining exactly how lighting could enhance theatrical elements for the audience. Despite this I believe this essay has already done it's best to critique itself as to ensure that the methods and theories mentioned have led to an effect on the audience such that the paper is still valuable.

References

Bloomsbury Media. (n.d.). *National Theatre Collection: Frankenstein - Learning Guide*. Bloomsbury Media. Retrieved January 3, 2023, from https://media.bloomsbury.com/rep/files/nt_frankenstein-learning-guide.pdf.

Bourn, J. (2010, December 5). *Colour Meaning: Meaning of the Color White*. Bourn Creative. Retrieved January 3, 2023, from https://www.bourncreative.com/meaning-of-the-color-white/

Brantley, B. (2011, February 24). Review of *It's (Gasp!) Alive, Not to Mention Peeved*. *The New York Times*. Retrieved January 3, 2023, from https://www.nytimes.com/2011/02/26/theater/reviews/26franken.html.

Clarerendon, E. (2020). REVIEW: Frankenstein, National Theatre at Home [web log]. Retrieved January 3, 2023, from https://lovelondonloveculture.com/2020/05/01/review-frankenstein-national-theatre-at-home/.

Dear, N. (2020). *Frankenstein (With Benedict Cumberbatch As The Creature)*. *National Theatre at Home* . Bloomberg Philantrophies. Retrieved January 1, 2023, from https://www.ntathome.com/videos/frankenstein-benedict-cumberbatch-as-the-creature.

DIGITTO MEDIA. (2019, August 29). *The importance of stage lighting*. American Audio Visual. Retrieved January 3, 2023, from https://www.americanaudiovisual.com/news/importance-stage-lighting/

Halliday, R. (2011, April). Frankenstein Lighting. *Theatrecrafts Lighting and Sound International* . Retrieved January 3, 2023, from https://www.theatrecrafts.com/archive/documents/lsi_2011april_frankenstein.pdf

McCandless, S. (1947). *A Method of Lighting the Stage* (3rd ed.). Reehl Litho Company.

Morton, J. L. (n.d.). Color & Vision Matters. Retrieved January 3, 2023, from https://www.colormatters.com/color-and-vision/color-and-vision-matters

Review of National Theatre's Frankenstein, *Lighting The Birth Of Frankenstein's Creature*. (2011, April 27). *etnow, entertainment technology news on web*. Retrieved January 3, 2023, from https://www.etnow.com/news/2011/4/lighting-the-birth-of-frankensteins-creature.

Salzberg, J. E., & Kupferman, J. (2009-2018). *Stage lighting for students: Principles of Composition, Functions of Stage Lighting, and Qualities of Light*. Stage Lighting

for Students. Retrieved January 3, 2023, from https://www.stagelightingprimer.com/index.html?slfs-functions.html&2

Salzberg, J. E., & Kupferman, J. (2009-2018). *Stage lighting for students: Types of Lighting Fixtures*. Stage Lighting for Students. Retrieved January 3, 2023, from http://www.stagelightingprimer.com/index.html?slfs-fixtures.html&2

Illuminated Integration. *Stage lighting 101 - everything you need to know*. Illuminated Integration. (2021, June 9). Retrieved January 3, 2023, from https://illuminated-integration.com/blog/stage-lighting-101/

Illuminated Integration. *Stage lighting: How to choose a color scheme*. Illuminated Integration. (2022, March 3). Retrieved January 3, 2023, from https://illuminated-integration.com/blog/stage-lighting-color-scheme/

theatretechgeek. (2020, May 19). Frankenstein – National Theatre Production [web log]. Retrieved January 3, 2023, from https://theatretechgeek.wordpress.com/2020/05/19/frankenstein-national-theatre-production/.

Trumbull, E. W. (2007, November 16). *The Audience and The Theatre*. Introduction to theatre online course. Retrieved January 3, 2023, from https://novaonline.nvcc.edu/eli/spd130et/audthea.htm

Watson, L. (1958). Color Concepts in Lighting Design. *Educational Theatre Journal*, *10*(3), 254–258. https://doi.org/10.2307/3216768

Wiegand, C. (2020, April 28). The National Theatre's Frankenstein: 'It was blinding – you felt the heat'. *The Guardian*. other. Retrieved January 3, 2023, from https://www.theguardian.com/stage/2020/apr/28/the-national-theatre-frankenstein-designer-mark-tildesley-danny-boyle.